AMITY UNIVERSITY PUNJAB, MOHALI

B.SC. MATHEMATICS (5TH SEMESTER)
SET THEORY AND METRIC SPACES (MAT304)

Unit I: Theory of Sets

Complete Lecture Notes

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Preface: Why Set Theory?

Set theory, founded by Georg Cantor in the late 19th century, forms the foundational language of modern mathematics. Every mathematical object—whether a number, a function, a geometric shape, or a topological space—can be defined as a set. This unit introduces the fundamental concepts of *cardinality* (the measure of "how many" elements a set has) and the *structure* of sets (orderings, functions, and relations). We will encounter the astonishing fact that there are *different sizes of infinity*, and we will explore the logical paradoxes that forced mathematicians to formalize set theory axiomatically (ZFC). The tools developed here—particularly Zorn's Lemma and the Axiom of Choice—are indispensable for higher mathematics, from algebra to analysis.

Lectures 1–2: Finite, Infinite, Countable, and Uncountable Sets

1.1 The Fundamental Question: "How Many?"

When we look at a finite set, we can count its elements. But how do we compare the "size" of infinite sets? The brilliant insight of Georg Cantor (1874) was: **Two sets have the same size if their elements can be paired up in a perfect one-to-one fashion (a bijection).**

Definition 0.1 (Bijection)**.** A bijection is a rule that matches every element of set A with exactly one element of set B, with no leftovers on either side. Think of it as a "perfect dance partner" system. If you can pair them up, they have the same *cardinality*.

Definition 0.2 (Cardinality)**.** The **cardinal number** (or cardinality) of a set A, denoted $|A|$, is the equivalence class of A under the relation "there exists a bijection between them." For finite sets, this aligns with the usual natural number.

1.2 Finite vs. Infinite Sets

- **Finite Set**: Can be matched with $\{1, 2, \ldots, n\}$ for some $n \in \mathbb{N}$. Examples: $\{a, b, c\}$, the set of planets.
- **Infinite Set**: Cannot be matched with any finite set. Examples: $\mathbb{N}, \mathbb{Z}, \mathbb{R}$.

Remark 0.1 (Dedekind's Characterization of Infinite Sets)**.** An infinite set is one that can be put into a one-to-one correspondence with a *proper subset* of itself.

Example 0.2. $\mathbb{N} = \{1, 2, 3, \ldots\}$ can be matched with the even positive integers $\{2, 4, 6, \ldots\}$ via the rule $f(n) = 2n$. This is a bijection between $\mathbb{N}$ and a *proper subset* of $\mathbb{N}$. This is **impossible** for finite sets. *Significance*: This immediately tells us that "part" can equal "whole" when dealing with infinity. This was historically shocking but is now the defining feature of infinite cardinalities.

1.3 Countable Sets (The "Smallest" Infinity)

Definition 0.3. A set is **countable** if it is finite or countably infinite (i.e., has a bijection with $\mathbb{N}$). We denote the cardinality of $\mathbb{N}$ by $\aleph_0$ (read "aleph-null").

Example 0.3 (Cantor's Pairing Function: $\mathbb{N} \times \mathbb{N}$ is Countable)**.** Define $f(m,n) = \frac{(m+n)(m+n+1)}{2} + n$. This is a bijection from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$. This proves that the Cartesian product of two countable sets is countable. *Intuition*: We can arrange the lattice points $\mathbb{N} \times \mathbb{N}$ in a spiral or diagonal list.

Example 0.4 (Hilbert's Hotel Analogy)**.** Imagine a hotel with infinitely many rooms (numbered $1, 2, 3, \ldots$).

- If a new guest arrives, move every existing guest from room n to room $n+1$, freeing room 1. ($\aleph_0 + 1 = \aleph_0$).
- If a bus with infinitely many new guests arrives, move the existing guest from room n to room $2n$, freeing all odd-numbered rooms. ($\aleph_0 + \aleph_0 = \aleph_0$).

Countable infinity is *absorbing* under addition and finite multiplication.

1.4 Key Theorems on Countability

Theorem 0.5 (Countable Union of Countable Sets is Countable)**.** The union of countably many countable sets is countable.

Proof (Cantor's Diagonal Argument in detail): Suppose we have countably many sets $A_1, A_2, A_3, \ldots$. List the elements of each A_i in a row:

$$\begin{aligned} A_1 &: a_{11}, a_{12}, a_{13}, \ldots \\ A_2 &: a_{21}, a_{22}, a_{23}, \ldots \\ A_3 &: a_{31}, a_{32}, a_{33}, \ldots \\ &\vdots \end{aligned}$$

Now traverse the array by *diagonals*:

$$a_{11}; \quad a_{12}, a_{21}; \quad a_{13}, a_{22}, a_{31}; \quad a_{14}, a_{23}, a_{32}, a_{41}; \ldots$$

Delete any duplicates. This creates a single infinite list containing every element from every A_i. Hence, the union is countable. *Note*: This proof assumes the Axiom of Choice for countably many sets (to choose an enumeration of each A_i), which is generally accepted without controversy.

Corollary 0.6. The set of integers $\mathbb{Z}$ is countable: $\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-\mathbb{N}\}$.

Corollary 0.7. The set of rational numbers $\mathbb{Q}$ is countable: For each denominator n, $Q_n = \{m/n : m \in \mathbb{Z}\}$ is countable. Since $\mathbb{Q} = \bigcup_{n=1}^{\infty} Q_n$, by the theorem, $\mathbb{Q}$ is countable. *Remark*: Despite being densely packed on the number line (between any two reals lies a rational), the rationals are just a "countable dust" compared to the reals. This distinguishes topological density from cardinal size.

Corollary 0.8. The set of real algebraic numbers is countable. An algebraic number is a root of a polynomial with integer coefficients. There are countably many polynomials (each degree n has $\aleph_0^{n+1} = \aleph_0$ coefficients), and each polynomial has finitely many roots. Therefore, the set of algebraic numbers is a countable union of finite sets.

1.5 Uncountable Sets: The Breakthrough

Theorem 0.9 (Cantor's Diagonal Argument, 1891)**.** The set $\mathbb{R}$ is uncountable. Specifically, the interval $(0, 1)$ is uncountable.

Proof (Detailed): Assume $(0, 1)$ is countable. Then we can list all its elements as infinite decimals:

$$\begin{aligned} a_1 &= 0.\mathbf{d_{11}}d_{12}d_{13}\dots \\ a_2 &= 0.d_{21}\mathbf{d_{22}}d_{23}\dots \\ a_3 &= 0.d_{31}d_{32}\mathbf{d_{33}}\dots \\ &\vdots \end{aligned}$$

We must be careful to avoid decimals ending in infinite 9s (e.g., 0.4999... = 0.5000...). We adopt the convention of using the non-terminating expansion (without trailing 9s). Construct a new number $b = 0.b_1b_2b_3\dots$ where:

$$b_n = \begin{cases} 1 & \text{if } d_{nn} \neq 1, \\ 2 & \text{if } d_{nn} = 1. \end{cases}$$

By construction, b differs from a_n in the n-th decimal place for every n. Therefore, b is not on the list. Contradiction.

Remark 0.10. This proves that there are "more" real numbers than integers. It introduces the **Cardinality of the Continuum**, denoted by $\mathfrak{c} = 2^{\aleph_0}$. *Immediate Consequence*: Since algebraic numbers are countable ($\aleph_0$) and reals are uncountable ($\mathfrak{c}$), the complement (transcendental numbers) has cardinality $\mathfrak{c}$. This is a pure existence proof—it tells us "hordes" of transcendentals exist without explicitly naming one (like π or e). This was a paradigm shift in mathematics: proving existence without construction.

Lectures 3–4: Cardinality and the Schröder–Bernstein Theorem

2.1 The Need for a "Comparison" Tool

Finding an explicit bijection between two sets is often difficult or tedious. The Schröder–Bernstein theorem provides a powerful shortcut: we only need to construct two *injections*.

Theorem 0.11 (Schröder–Bernstein (Cantor–Bernstein))**.** If there exists an injective map $f : A \to B$ (i.e., $|A| \leq |B|$) and an injective map $g : B \to A$ (i.e., $|B| \leq |A|$), then there exists a bijection $h : A \to B$. Thus $|A| = |B|$.

Proof Sketch (Kaplansky's Cycle Decomposition): Take $F = g \circ f : A \to A$, which is an injection from A into itself. We decompose A into disjoint *chains* or *cycles* under the map F:

- **Finite cycles**: $a_1 \to a_2 \to \cdots \to a_n \to a_1$.
- **Bilateral infinite chains**: $\cdots \to a_{-2} \to a_{-1} \to a_0 \to a_1 \to \cdots$ (indexed by $\mathbb{Z}$).
- **Unilateral infinite chains**: $a_1 \to a_2 \to a_3 \to \cdots$ (with a starting point not in the image of F).

Since $C = g(B)$ lies between $f(A)$ and A, C contains $f(A)$ and is contained in A. On finite and bilateral cycles, F is already a bijection, so we leave them unchanged or take the identity. On unilateral chains, C either includes the starting point or not. We define h to be F (which shifts everything forward) on the chains where the starting point is missing from C, and the identity on the chains where it is included. This yields a bijection between A and C. Since $g : B \to C$ is already a bijection (as g is injective and $C = g(B)$), composing gives a bijection $A \to B$.

2.2 Stunning Applications of Schröder–Bernstein

Example 0.12. Comparing $(0, 1)$ **and** $[0, 1]$:

- Injection $f : (0, 1) \to [0, 1]$: $f(x) = x$ (inclusion).
- Injection $g : [0, 1] \to (0, 1)$: $g(x) = \frac{x}{2} + \frac{1}{4}$. This maps $0 \mapsto 1/4$ and $1 \mapsto 3/4$, keeping everything strictly inside.
- By S-B, $|(0, 1)| = |[0, 1]|$. Adding two points to an uncountable interval does not change its cardinality!

Example 0.13. The Line and the Plane $(|\mathbb{R}| = |\mathbb{R}^2|)$:

- Injection $\mathbb{R} \to \mathbb{R}^2$: $x \mapsto (x, 0)$.
- Injection $\mathbb{R}^2 \to \mathbb{R}$: Write $x = 0.a_1a_2a_3\ldots$ and $y = 0.b_1b_2b_3\ldots$ (using non-terminating expansions). Map to $0.a_1b_1a_2b_2a_3b_3\ldots$. This is injective (though not surjective).
- By S-B, the line and the plane have the **exact same number of points**. Cantor famously wrote, "I see it, but I don't believe it."

Example 0.14. Cardinality of Function Spaces: Let $C(\mathbb{R}, \mathbb{R})$ be the set of continuous functions from $\mathbb{R}$ to $\mathbb{R}$.

- A continuous function is determined by its values on $\mathbb{Q}$ (which is countable). Thus $|C(\mathbb{R}, \mathbb{R})| \leq |\mathbb{R}^{\mathbb{Q}}| = \mathfrak{c}^{\aleph_0} = \mathfrak{c}$.
- The constant functions provide an injection $\mathbb{R} \hookrightarrow C(\mathbb{R}, \mathbb{R})$.
- By S-B, $|C(\mathbb{R}, \mathbb{R})| = \mathfrak{c}$.
- In contrast, the set of *all* functions $\mathbb{R} \to \mathbb{R}$ has cardinality $\mathfrak{c}^{\mathfrak{c}}$, which is strictly larger than $\mathfrak{c}$ by Cantor's theorem.

Lectures 5–6: Cantor's Theorem and Order Relation

3.1 Cantor's Theorem: The Diagonal Principle

Theorem 0.15 (Cantor's Theorem)**.** For every set A, $|A| < |P(A)|$. There is no surjection from A onto its power set.

Proof (Detailed): Assume for contradiction that $f : A \to P(A)$ is surjective. Define the set $D = \{a \in A : a \notin f(a)\}$. Since f is onto, there exists $b \in A$ such that $f(b) = D$. Now ask: Is $b \in D$?

- If $b \in D$, then by definition of D, $b \notin f(b) = D$, a contradiction.
- If $b \notin D$, then by definition of D, $b \in f(b) = D$, a contradiction.

Thus, no such surjection exists. Since the map $a \mapsto \{a\}$ is an injection from A into $P(A)$, we have $|A| < |P(A)|$.

Example 0.16. Finite Case Check: For $A = \{1, 2\}$, $|A| = 2$, $|P(A)| = 4$. Indeed $2 < 4$. Cantor's theorem is the infinite generalization of this simple fact.

Remark 0.17. This theorem creates an infinite, strictly increasing hierarchy of cardinal numbers:

$$\aleph_0 < 2^{\aleph_0} = \mathfrak{c} < 2^{\mathfrak{c}} < 2^{2^{\mathfrak{c}}} < \cdots$$

There is no "largest" cardinal number. This implies that the class of all cardinal numbers is a proper class (too large to be a set).

3.2 Ordering Cardinal Numbers

We define $\alpha \leq \beta$ if there exists an injection $A \hookrightarrow B$.

- **Reflexive**: The identity map is an injection.
- **Antisymmetric**: This is precisely the Schröder–Bernstein theorem.
- **Transitive**: The composition of two injections is an injection.

Thus, the cardinal numbers form a *partial order*. However, a much stronger statement holds:

Theorem 0.18 (Zermelo's Comparability Theorem)**.** For any two sets A and B, either $|A| \leq |B|$ or $|B| \leq |A|$ (or both). In other words, cardinal numbers form a *total order* (a chain).

Why this is deep: This theorem is equivalent to the Axiom of Choice. Without AC, there might be two sets whose cardinalities are incomparable. We will prove this using Zorn's Lemma in Lectures 9-11.

Lectures 7–8: Arithmetic of Cardinal Numbers

4.1 Cardinal Addition

For cardinals $\alpha = |A|$, $\beta = |B|$, define $\alpha + \beta = |A \sqcup B|$ (the disjoint union).

Theorem 0.19 (Absorption Law for Addition)**.** If α, β are cardinals and at least one is infinite, then $\alpha + \beta = \max\{\alpha, \beta\}$.

Proof Idea: Suppose $\alpha \leq \beta$, β infinite. We have $\beta \leq \alpha + \beta \leq \beta + \beta$. It can be shown (using a bijection between $\beta \times \{0, 1\}$ and β) that $\beta + \beta = \beta$. Therefore $\alpha + \beta = \beta$.

Example 0.20. $\aleph_0 + \mathfrak{c} = \mathfrak{c}$. Hilbert's Hotel: filling countably many new guests into an uncountable hotel leaves the uncountable hotel unchanged.

4.2 Cardinal Multiplication

Define $\alpha \cdot \beta = |A \times B|$.

Theorem 0.21 (Absorption Law for Multiplication)**.** If α, β are infinite and non-zero, then $\alpha \cdot \beta = \max\{\alpha, \beta\}$.

Proof Idea: Requires the theorem $\alpha \cdot \alpha = \alpha$ for any infinite cardinal α. This is proved via Zorn's Lemma by taking a maximal subset $E \subset A$ such that $|E \times E| = |E|$. A maximal E must have $|A \setminus E| < |E|$, forcing equality.

Example 0.22. $\mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$. "The line has as many points as the plane."

4.3 Cardinal Exponentiation (The Rich Structure)

Define $\alpha^\beta = |\{f : B \to A\}|$, where $|A| = \alpha$, $|B| = \beta$. This is the number of functions from a set of size β to a set of size α.

Theorem 0.23 (Fundamental Identity)**.** $2^{\aleph_0} = \mathfrak{c}$. (The power set of $\mathbb{N}$ has the same cardinality as $\mathbb{R}$).

Proof: - $2^{\aleph_0} \leq \mathfrak{c}$: Each subset of $\mathbb{N}$ corresponds to a binary sequence, which maps to a real number in $[0, 1]$ via binary expansion. - $\mathfrak{c} \leq 2^{\aleph_0}$: Each real number in $(0, 1)$ has a unique binary expansion (avoiding trailing 1s), giving a subset of $\mathbb{N}$.

Theorem 0.24 (Exponentiation Laws)**.** For all cardinals α, β, γ:

1. $(\alpha \cdot \beta)^\gamma = \alpha^\gamma \cdot \beta^\gamma$.
2. $\alpha^{\beta+\gamma} = \alpha^\beta \cdot \alpha^\gamma$.
3. $(\alpha^\beta)^\gamma = \alpha^{\beta \cdot \gamma}$.

Proof via Function Spaces: 1. A function from C to $A \times B$ is a pair of functions (f, g) where $f : C \to A$ and $g : C \to B$. 2. A function from $B \sqcup C$ to A is a pair of functions (f, g) where $f : B \to A$ and $g : C \to A$. 3. A function from $B \times C$ to A corresponds to a function from C to A^B via currying: $f(b, c) = g(c)(b)$.

Example 0.25. $\mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0} = \mathfrak{c}$. The number of *sequences* of real numbers is the same as the number of real numbers!

Example 0.26. $\mathfrak{c}^\mathfrak{c} = (2^{\aleph_0})^\mathfrak{c} = 2^{\aleph_0 \cdot \mathfrak{c}} = 2^\mathfrak{c}$. The number of functions from $\mathbb{R}$ to $\mathbb{R}$ is strictly larger than $\mathfrak{c}$.

4.4 The Continuum Problem (Hilbert's First Problem)

Where does $\mathfrak{c}$ fit in the aleph hierarchy? Cantor conjectured that $\mathfrak{c} = \aleph_1$. This is the **Continuum Hypothesis (CH)**.

- **Gödel (1940)**: CH is **consistent** with ZFC (cannot be disproved). He constructed the inner model L (constructible universe) where CH holds.
- **Cohen (1963)**: CH is **independent** of ZFC (cannot be proved). He invented the method of *forcing* to construct models where CH fails (e.g., where $2^{\aleph_0} = \aleph_2$).

Conclusion: The size of the continuum is not determined by the standard axioms of mathematics. It is a "free parameter" in the universe of sets.

Lectures 9–11: Zorn's Lemma and the Axiom of Choice

5.1 The Axiom of Choice (AC)

Statement (AC): For any collection $\{A_i\}_{i\in I}$ of non-empty sets, there exists a *choice function* $f : I \to \bigcup A_i$ such that $f(i) \in A_i$ for all i. Equivalently, the Cartesian product $\prod_{i\in I} A_i$ is non-empty.

Remark 0.27. AC seems trivially true for finite collections or when the sets have a definite rule (e.g., "choose the smallest element" from subsets of $\mathbb{N}$). The controversy arises for arbitrary, uncountable collections of sets where no such rule exists. AC is non-constructive.

5.2 Zorn's Lemma (ZL)

Statement: Let L be a non-empty partially ordered set. If every chain (totally ordered subset) in L has an upper bound in L, then L contains at least one maximal element.

Example 0.28 (Intuition: The Mountain Climbing Analogy)**.** Imagine you are climbing a mountain range. If you are not at a peak, you can climb higher. If you have a chain of climbers stretching infinitely, the premise guarantees a "plateau" or peak above them. Zorn's lemma says you will eventually find a peak (maximal element).

5.3 The Equivalence (AC $\leftrightarrow$ ZL $\leftrightarrow$ WO)

Theorem 0.29 (Zermelo)**.** Over ZF set theory, the following are equivalent:

1. The Axiom of Choice (AC).
2. Zorn's Lemma (ZL).
3. The Well-Ordering Theorem (WO): Every set can be well-ordered (i.e., every non-empty subset has a least element).

Proof Sketches: - **AC $\Rightarrow$ ZL**: Suppose L has no maximal element. Then for every $x \in L$, the set $M_x = \{y \in L : y > x\}$ is non-empty. Using AC, choose $g(x) \in M_x$. Thus $g(x) > x$. One then constructs a strictly increasing chain indexed by ordinals (transfinite recursion) longer than L, a contradiction. (This uses the fact that ordinals form a proper class). - **ZL $\Rightarrow$ WO**: Given a set S, consider the poset W of all well-ordered subsets of S, ordered by "is an initial segment of". Every chain has an upper bound (the union). ZL gives a maximal well-ordered subset. If it missed an element s, we could append s to the end, contradicting maximality. Hence it is S. - **WO $\Rightarrow$ AC**: Well-order the union $\bigcup A_i$. For each set A_i, define $f(A_i)$ to be the least element of A_i under the well-order.

5.4 Detailed Applications of Zorn's Lemma

Example 0.30. Every Vector Space has a Basis (Crucial for Functional Analysis): Let V be a vector space over a field K. Let L be the set of all linearly independent subsets of V, ordered by inclusion.

- *Chain Check*: Let $\{B_i\}$ be a chain. The union $B = \cup B_i$ is linearly independent. If a finite linear combination of elements in B sums to zero, all those elements lie in some single B_i (because the set of coefficients is finite and the chain is totally ordered). This contradicts the independence of B_i. Thus B is an upper bound.
- ZL gives a maximal independent set B.
- If B did not span V, there would be a vector $v \notin \text{span}(B)$. Then $B \cup \{v\}$ is linearly independent, contradicting maximality.

Significance: This proves that every Hilbert space, every function space, and every abstract vector space has a basis, allowing the definition of dimension for infinite-dimensional spaces.

Example 0.31. Existence of Maximal Ideals in Commutative Rings: Let R be a ring with unity. Let L be the set of all proper ideals (ideals not containing 1), ordered by inclusion.

- *Chain Check*: Union of a chain of proper ideals is proper. If 1 were in the union, it would be in some ideal in the chain, contradiction.
- ZL yields a maximal ideal M.
- *Significance*: The quotient R/M is a field. This is foundational for algebraic geometry and modern number theory (e.g., proving the existence of prime ideals).

Example 0.32. Comparability of Cardinals (Proof of Total Order): Given two sets A and B. Consider the poset L of all triples (A', B', f) where $A' \subset A$, $B' \subset B$, and $f : A' \to B'$ is a bijection. Order by extension.

- *Chain Check*: Union of a chain of bijections gives a bijection on the union of domains.
- ZL gives a maximal triple (A_0, B_0, f_0).
- If $A_0 \neq A$ and $B_0 \neq B$, pick $a \in A \setminus A_0$ and $b \in B \setminus B_0$. Extend f_0 by $f(a) = b$, contradicting maximality.
- Hence either $A_0 = A$ (so A injects into B) or $B_0 = B$ (so B injects into A).

5.5 The Controversial Nature of AC

While AC is widely accepted in modern mathematics, it has bizarre consequences:

- **Banach-Tarski Paradox**: A solid ball in $\mathbb{R}^3$ can be decomposed into finitely many pieces and reassembled into two identical copies of the original ball. This relies heavily on AC.
- **Non-measurable Sets**: AC implies the existence of subsets of $\mathbb{R}$ (Vitali sets) that are not Lebesgue measurable.

However, rejecting AC also leads to pathologies (e.g., a vector space without a basis, an infinite set that is not Dedekind-infinite). Most mathematicians accept ZFC (Zermelo-Fraenkel with Choice) as the standard foundation.

Lecture 12: Various Set-Theoretic Paradoxes

6.1 The Crisis in Foundations

At the turn of the 20th century, Cantor's naive set theory (based on the "comprehension principle": given a property ϕ, $\{x : \phi(x)\}$ is a set) was discovered to be inconsistent. These paradoxes showed that not every collection is a set.

6.2 Russell's Paradox (1901)

Define $R = \{X \mid X \text{ is a set and } X \notin X\}$.

- If $R \in R$, then by definition $R \notin R$, contradiction.
- If $R \notin R$, then by definition $R \in R$, contradiction.

Verdict: R cannot be a set. It is a **proper class**. The problem is the self-referential nature of the definition.

Example 0.33 (Grelling's Paradox (Linguistic Version))**.** An adjective is *heterological* if it does not describe itself (e.g., "long" is not long). Is "heterological" heterological? This shows the paradox is not confined to mathematics but arises in natural language.

6.3 Burali-Forti Paradox (1897)

Let Ω be the set of all ordinal numbers. Ω is well-ordered, hence has an ordinal α. But then $\alpha \in \Omega$, and Ω is order-isomorphic to the initial segment of ordinals less than α, contradicting the fact that no well-ordered set is isomorphic to a proper initial segment of itself. **Verdict**: The collection of all ordinals is a proper class.

6.4 Cantor's Paradox (1899)

If V is the set of all sets, then $|V| < |P(V)|$ by Cantor's theorem. But $P(V) \subset V$, so $|P(V)| \leq |V|$. Contradiction. **Verdict**: There is no universal set. The class of all sets is a proper class.

6.5 How ZFC Axioms Avoid These Pitfalls

- **Axiom of Separation (Subset Axiom)**: You cannot form $\{x : \phi(x)\}$. You can only form $\{x \in A : \phi(x)\}$ for a pre-existing set A. To form Russell's set R, you would need a universal set A to start from, which does not exist.
- **Axiom of Foundation (Regularity)**: Every non-empty set A contains an element x such that $x \cap A = \emptyset$. This rules out self-membership (no set is an element of itself).
- **Axiom of Replacement**: Controls the size of images of functions, preventing the formation of "too large" sets (like the set of all ordinals).

These axioms ensure that we only build sets in a cumulative hierarchy (the von Neumann universe V) starting from $\emptyset$, step by step through ordinals, avoiding any circularity.

Lectures 13–15: Tutorial and Problem-Solving Session

7.1 Session 13: Deep Conceptual Questions

Objective: Understand the "why" behind the definitions.

1. **Question**: "A proper subset can have the same cardinality as the whole set." Give three distinct examples (one countable, one uncountable, one for an arbitrary cardinal κ).

2. **Question**: "The rationals are countable, yet dense in the reals." Explain why this is not a contradiction. What is the topological versus the set-theoretic notion of size?

3. **Question**: "Without the Axiom of Choice, mathematics looks very different." Discuss the possible pathologies of rejecting AC. (Hint: Consider vector spaces and the comparability of cardinals.)

4. **Question**: Why does Cantor's diagonal argument fail if we try to apply it to the rational numbers? (Answer: The constructed decimal b might not be rational.)

7.2 Session 14: Core Computational Problems

Objective: Master the mechanics of bijections and cardinal arithmetic.

1. Prove that $\mathbb{N} \times \mathbb{N}$ is countable using the Cantor pairing function. *Hint*: Show $f(m,n) = \frac{(m+n)(m+n+1)}{2} + n$ is a bijection.

2. Prove that the set of all polynomials with rational coefficients is countable. *Hint*: $\mathbb{Q}[x] = \bigcup_{n=0}^{\infty} \mathbb{Q}^n$.

3. Show $|(0,1)| = |\mathbb{R}|$ using the explicit map $f(x) = \tan(\pi x - \pi/2)$. (Assume the domain $(-\pi/2, \pi/2)$).

4. Find injections both ways between $\mathbb{R}$ and $\mathbb{R} \times \mathbb{R}$ using interleaving decimals. *Hint*: $0.a_1a_2\ldots$ and $0.b_1b_2\ldots \mapsto 0.a_1b_1a_2b_2\ldots$.

5. Compute the following cardinalities:

   (a) $\mathfrak{c}^2$

   (b) $\mathfrak{c}^{\aleph_0}$

   (c) $(\mathfrak{c}^{\mathfrak{c}})^{\aleph_0}$

   (d) $\aleph_0^{\mathfrak{c}}$ (Answer: $2^{\mathfrak{c}}$)

6. Prove that the set of finite subsets of $\mathbb{N}$ is countable, while the set of all subsets is uncountable.

7. Show that $\mathbb{R}$ is uncountable by showing that $(0,1)$ is uncountable.

7.3 Session 15: Advanced and Theoretical Problems

Objective: Apply Zorn's Lemma and understand the logical foundations.

1. **(Zorn's Lemma)**: Prove that in any partially ordered set, every chain is contained in a maximal chain.

2. **(Zorn's Lemma)**: Prove Zorn's lemma implies the Well-Ordering theorem. *Hint*: Consider well-ordered subsets of a given set S, ordered by initial segment extension.

3. **(Paradoxes)**: Write a short note explaining why the "barber paradox" is not a valid mathematical contradiction but Russell's paradox is, and how the Axiom of Separation resolves the latter.

4. **(Cantor-Bendixson, easy version)**: Prove that every closed subset of $\mathbb{R}$ is either countable or has cardinality $\mathfrak{c}$. *Hint*: Use the fact that a perfect set (closed with no isolated points) contains a Cantor set, which has cardinality $\mathfrak{c}$.

5. **(Cardinal Arithmetic)**: Suppose κ is an infinite cardinal. Prove that $\kappa^{\aleph_0} = \max\{\kappa, 2^{\aleph_0}\}$. *Hint*: $\kappa^{\aleph_0} \leq (2^\kappa)^{\aleph_0} = 2^{\kappa \cdot \aleph_0} = 2^\kappa$, but that's too large. Use the fact that if $\kappa \leq 2^{\aleph_0}$, equality to $2^{\aleph_0}$; if $\kappa > 2^{\aleph_0}$, it requires more advanced methods (König's theorem) to show it's not trivial.

6. **(Application of S-B)**: Prove that the set of all sequences of real numbers (i.e., $\mathbb{R}^{\mathbb{N}}$) has cardinality $\mathfrak{c}$.

7. **(Equivalence Classes)**: Let $\sim$ be an equivalence relation on a set A. Prove that the collection of equivalence classes partitions A. Show that the map sending each element to its equivalence class is a surjection from A onto the set of classes.

7.4 Detailed Hints for Advanced Problems

- **For Problem 1 (Maximal Chain)**: Let L be the poset of all chains in your original poset, ordered by inclusion. Show that the union of a chain of chains is a chain. Apply Zorn's Lemma.

- **For Problem 4 (Cantor-Bendixson)**: The key lemma is that a perfect subset of $\mathbb{R}$ contains a Cantor set. You can construct it by repeatedly removing intervals, similar to the ternary Cantor set construction. The remaining set is uncountable and has no isolated points, hence has cardinality $\mathfrak{c}$.

- **For Problem 6 (Sequences of Reals)**: $\mathbb{R}^{\mathbb{N}} = (\mathbb{R})^{\aleph_0} = \mathfrak{c}^{\aleph_0} = \mathfrak{c}$.

Final Summary Table of Significance

Concept	Intuition	Consequence / Significance
Countable Infinity	The smallest infinity. Can be "listed" in an infinite queue.	$\mathbb{Q}$ is countable. Algebraic numbers are countable.
Uncountable Infinity	Cannot be listed. Any list misses something.	$\mathbb{R}$ is uncountable. Existence of transcendentals.
Schröder-Bernstein	No need for explicit bijection; two injections suffice.	Proves line has as many points as the plane. Fundamental for comparing infinities.
Cantor's Theorem	Power sets are strictly larger.	Infinite ladder of infinities. No largest cardinal. Proves "set of all sets" is impossible.
Cardinal Arithmetic	Addition/Multiplication are absorbing; Exponentiation creates new sizes.	$2^{\aleph_0} = \mathfrak{c}$. The continuum problem (CH) is independent.
Zorn's Lemma / AC	Allows infinite, non-constructive choices.	Every vector space has a basis. Maximal ideals. Equivalent to Well-Ordering.
Paradoxes	Naive set theory is inconsistent.	Necessitated the ZFC Axioms. Separates "sets" from "proper classes".

End of Unit I Notes